

OBJECT ORIENTED PROGRAMMING- PROJECT

Project Name : Restaurant Management

Team members : 3

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Objectives:

- To develop a simple Restaurant Management system using C++.
- To implement Object-Oriented Programming concepts such as classes and inheritance.
- To perform table booking and management using arrays.
- To enable food ordering and bill calculation.
- To apply file handling for storing customer records permanently.
- To demonstrate exception handling for invalid inputs.
- To implement basic user interaction using standard I/O (input/output) functions.

Description:

This program is a simple Restaurant management system developed using C++. It allows users to enter their details, book a table, order food, and generate a bill. The program checks table availability, calculates the total amount based on selected items, and ensures valid payment. Customer details and order information are stored in a file for future reference.

The program uses object-oriented concepts like

- classes & object
- Constructor
- Inheritance (Single inheritance)
- Function implementation
- file handling (To store customer and booking details permanently in a file.)
- exception handling (Handling runtime errors -to handle invalid table selection)

for efficient operation.

Source Code:

Restaurant Name: SEBYEOL RESTAURANT!!

```
#include<iostream>
#include<fstream>
#include<sstream>
#include<string>
using namespace std;
class hotel{
public:
    char name[30];
    long int mno;
```

```

int amount, tableno, time, pay;
int tables[10];

hotel(){
    amount = 0;
    int temp[10] = {1,0,0,0,1,1,0,0,1,1};
    for(int i=0; i<10; i++){
        tables[i] = temp[i];
    }
}

};
class booking : public hotel{
public:
    string items = "";
    void getdata(){
        cout<<"Enter your name: ";
        cin.getline(name, 30);
        cout<<"Mobile number: ";
        cin>>mno;
        cout<<"Enter booking time (0-23): ";
        cin>>time;

        if(time < 0 || time > 23){
            cout<<"Invalid time! Setting default 12.\n";
            time = 12;
        }
    }
    void book();
    void orderFood();
    void generateBill();
    void saveToFile();
};
void booking::book(){
    cout<<"\nAvailable Tables:\n";
    for(int i=0; i<10; i++){
        if(tables[i] == 0){
            cout<<"Table No: "<<i+1<<endl;
        }
    }
}
while(true){

```

```

cout<<"Enter table number: ";
cin>>tableno;
try {
    if (tableno < 1 || tableno > 10) {
        throw "Invalid table number!";
    }
    if (tables[tableno - 1] == 1) {
        throw "Table already booked! Try another.";
    }
    tables[tableno - 1] = 1;
    cout << "Table booked successfully!\n";
    break;
}
catch (const char* msg) {
    cout << msg << endl;
}
}
}

void booking::orderFood(){
    int n, choice, qty;

    cout<<"\nHow many items do you want to order? ";
    cin>>n;
    for(int i=0; i<n; i++){
        cout<<"\n---- MENU ----\n";
        cout<<"1. Pizza - 200\n";
        cout<<"2. Burger - 100\n";
        cout<<"3. Pasta - 150\n";
        cout<<"4. Coffee - 50\n";
        cout<<"Enter choice: ";
        cin>>choice;
        cout<<"Enter quantity: ";
        cin>>qty;
        stringstream ss;
        ss << qty;
        switch(choice){
            case 1:
                amount += 200 * qty;
                items += "Pizza x" + ss.str() + "\n";
                break;

```

```

        case 2:
            amount += 100 * qty;
            items += "Burger x" + ss.str() + "\n";
            break;

        case 3:
            amount += 150 * qty;
            items += "Pasta x" + ss.str() + "\n";
            break;

        case 4:
            amount += 50 * qty;
            items += "Coffee x" + ss.str() + "\n";
            break;

        default:
            cout<<"Invalid choice! Try again.\n";
            i--; // retry
    }
}

void booking::generateBill(){
    cout<<"                SEBYEOL RESTAURANT                \n";
    cout<<"\n----- BILL ----- \n";
    cout<<"Name: "<<name<<endl;
    cout<<"Mobile: "<<mno<<endl;
    cout<<"Time: "<<time<<endl;
    cout<<"Table No: "<<tableno<<endl;
    cout<<"\nItems Ordered:\n"<<items;
    cout<<"Total Amount: Rs."<<amount<<endl;
    do{
        cout<<"Enter payment: ";
        cin>>pay;
        if(pay < amount){
            cout<<"Insufficient payment! Try again.\n";
        }
    } while(pay < amount);

    cout<<"Payment Successful!\n";
}

```

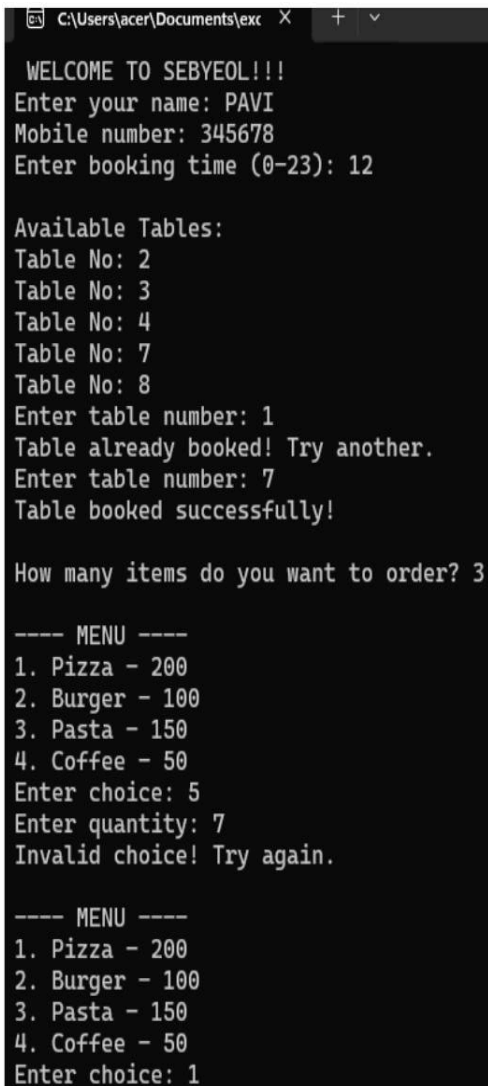
```

    cout<<"Balance: "<<pay - amount<<endl;
}
void booking::saveToFile(){
    ofstream file("hotel_records.txt", ios::app);
    if(!file){
        cout<<"Error opening file!\n";
        return;
    }
    file<<"----- CUSTOMER RECORD ----- \n";
    file<<"Name: "<<name<<endl;
    file<<"Mobile: "<<mno<<endl;
    file<<"Time: "<<time<<endl;
    file<<"Table No: "<<tableno<<endl;
    file<<"Items Ordered:\n"<<items;
    file<<"Total Amount: Rs."<<amount<<endl;
    file<<"----- \n\n";
    file.close();
    switch(tableno){
        case 2:
            cout<<"Verify Code: BYL2781";
            break;
        case 3:
            cout<<"Verify Code: BYL2782";
            break;
        case 4:
            cout<<"Verify Code: BYL2783";
            break;
        case 7:
            cout<<"Verify Code: BYL2784";
            break;
        case 8:
            cout<<"Verify Code: BYL2785";
            break;
        default:
            cout<<"Verify Code: BYL0000";
    }
    cout<<"\nData saved to file successfully!\n";
}
int main(){
    cout<<"                WELCOME TO SEBYEOL!!!                \n";

```

```
    booking b;  
    b.getdata();  
    b.book();  
    b.orderFood();  
    b.generateBill();  
    b.saveToFile();  
    return 0;  
}
```

Output:



```
WELCOME TO SEBYEOL!!!  
Enter your name: PAVI  
Mobile number: 345678  
Enter booking time (0-23): 12  
  
Available Tables:  
Table No: 2  
Table No: 3  
Table No: 4  
Table No: 7  
Table No: 8  
Enter table number: 1  
Table already booked! Try another.  
Enter table number: 7  
Table booked successfully!  
  
How many items do you want to order? 3  
  
---- MENU ----  
1. Pizza - 200  
2. Burger - 100  
3. Pasta - 150  
4. Coffee - 50  
Enter choice: 5  
Enter quantity: 7  
Invalid choice! Try again.  
  
---- MENU ----  
1. Pizza - 200  
2. Burger - 100  
3. Pasta - 150  
4. Coffee - 50  
Enter choice: 1
```

```
C:\Users\acer\Documents\exc X + v
4. Coffee - 50
Enter choice: 1
Enter quantity: 3

---- MENU ----
1. Pizza - 200
2. Burger - 100
3. Pasta - 150
4. Coffee - 50
Enter choice: 3
Enter quantity: 2

---- MENU ----
1. Pizza - 200
2. Burger - 100
3. Pasta - 150
4. Coffee - 50
Enter choice: 2
Enter quantity: 2
  SEBYEOL RESTAURANT

----- BILL -----
Name: PAVI
Mobile: 345678
Time: 12
Table No: 7

Items Ordered:
Pizza x3
Pasta x2
```

```
C:\Users\acer\Documents\exc X + v

---- MENU ----
1. Pizza - 200
2. Burger - 100
3. Pasta - 150
4. Coffee - 50
Enter choice: 2
Enter quantity: 2
SEBYEOL RESTAURANT

----- BILL -----
Name: PAVI
Mobile: 345678
Time: 12
Table No: 7

Items Ordered:
Pizza x3
Pasta x2
Burger x2
Total Amount: Rs.1100
Enter payment: 1100
Payment Successful!
Balance: 0
Verify Code: BYL2784
Data saved to file successfully!

-----
Process exited after 97.01 seconds with return value 0
Press any key to continue . . . |
```
